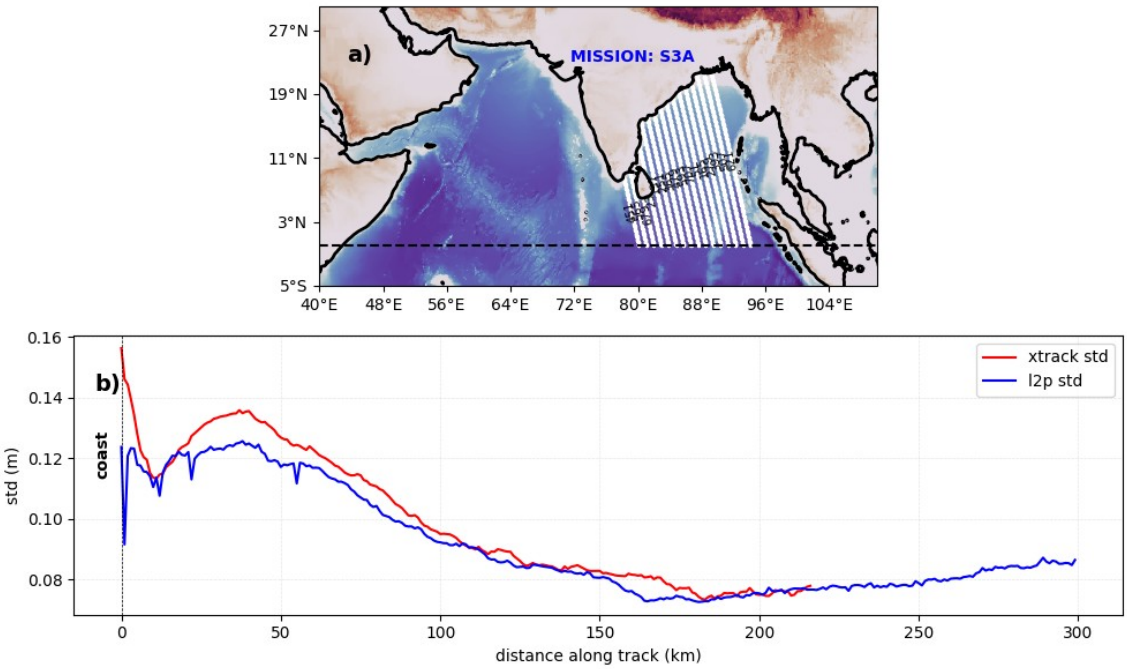


Fig1: Variability of sea level anomaly near the coast: b) Mean standard deviation of SLA of the tracks meeting the east coast of India for the satellite mission S3A compared between Xtrack along track products(red) and AVISO l2p tracks(blue). x-axis shows distance away from the coast in km a) shows the tracks from which mean standard deviation is calculated. The tracks chosen are common for both l2p and Xtrack products.

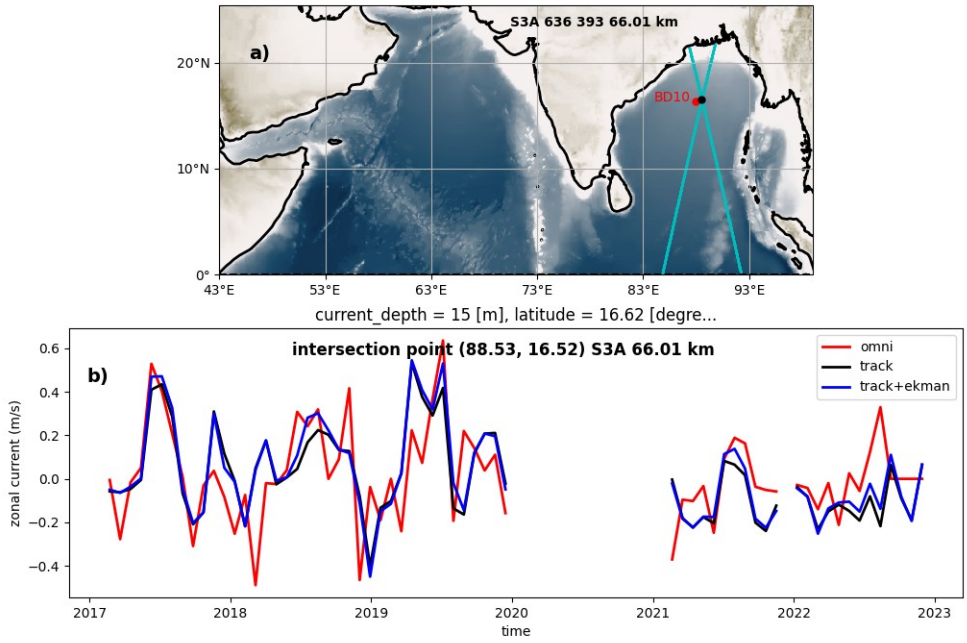


Fig2: b) Time series of geostrophic zonal component (black) derived at an intersection point which is close to an OMNI buoy (BD10) overlied with buoy's zonal current(red). Timeseries of gu correted with ekman u component is also added (blue). a) shows the intersection point(black) of the tracks 636 and 393 and close by bouy(red). It is rougly half a degree away from the intersection point.

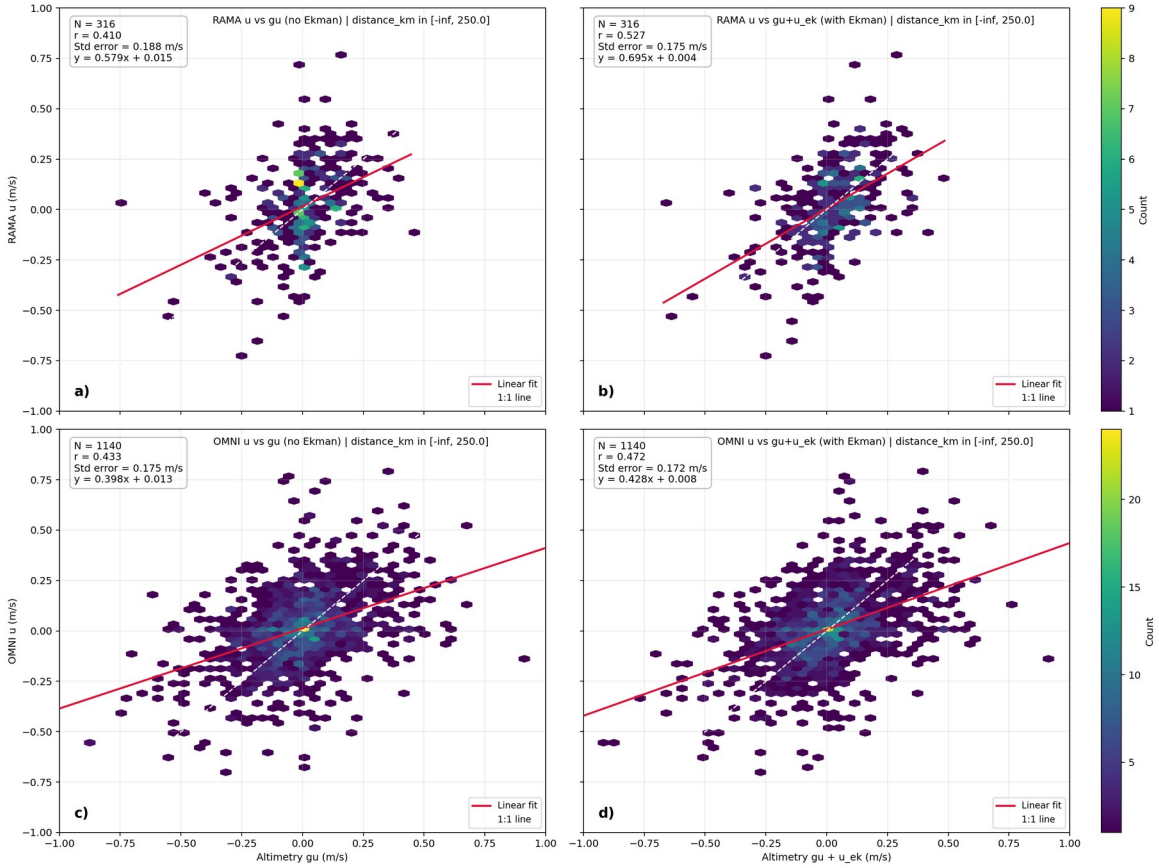


Fig3: a) Scatter plot of rama buoys' zonal currents(y-axis) versus altimetry derived zonal currents(x-axis) at nearby intersection points. b) Shows scatter plot with altimetry derived currents corrected with ekman zonal currents. c) and d) show scatter plots of OMNI currents with and without Ekman correction against altimetry zonal currents. The colorbar shows density of points in the scatter plot in terms of count. A redline passing through the points indicates linear fit.

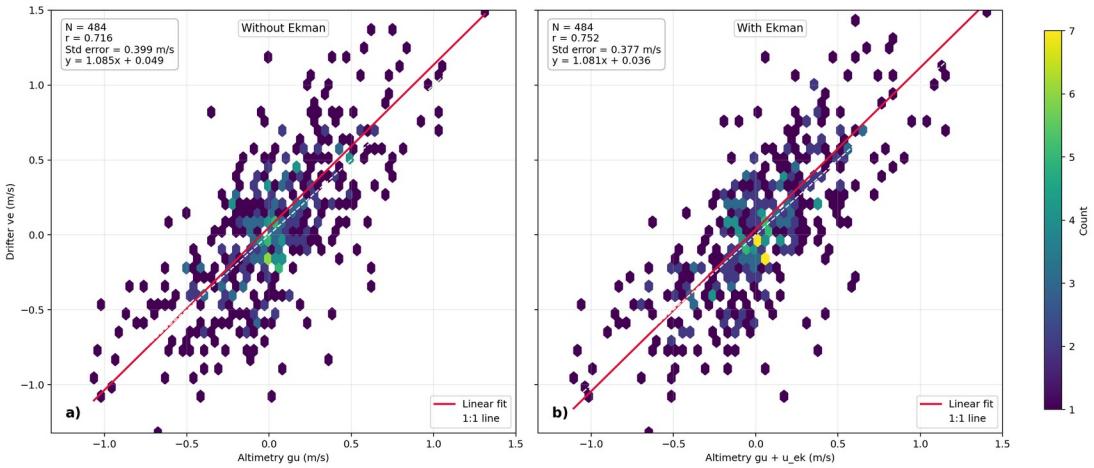


Fig4: a) Scatter plot of drifters' zonal velocities versus altimetry derived velocities within 200km inshore of the coast for the satellite TP+J1+J2+J3+S6A. b) shows scatter plot with altimetry derived velocities corrected with ekman zonal velocities.

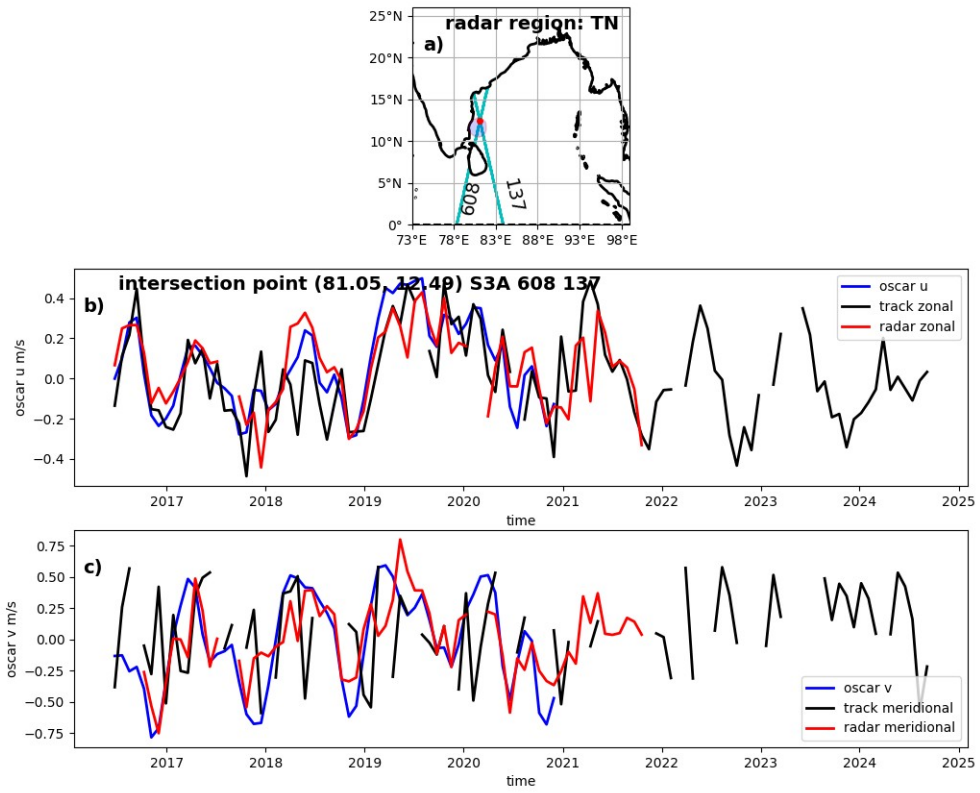


Fig5: a) Shows the radar region of TN with a intersection point within the region. b) and c) show zonal and meridional currents of altimetry derived geostrophic currents at the intersection point along with radar zonal velocity time series.

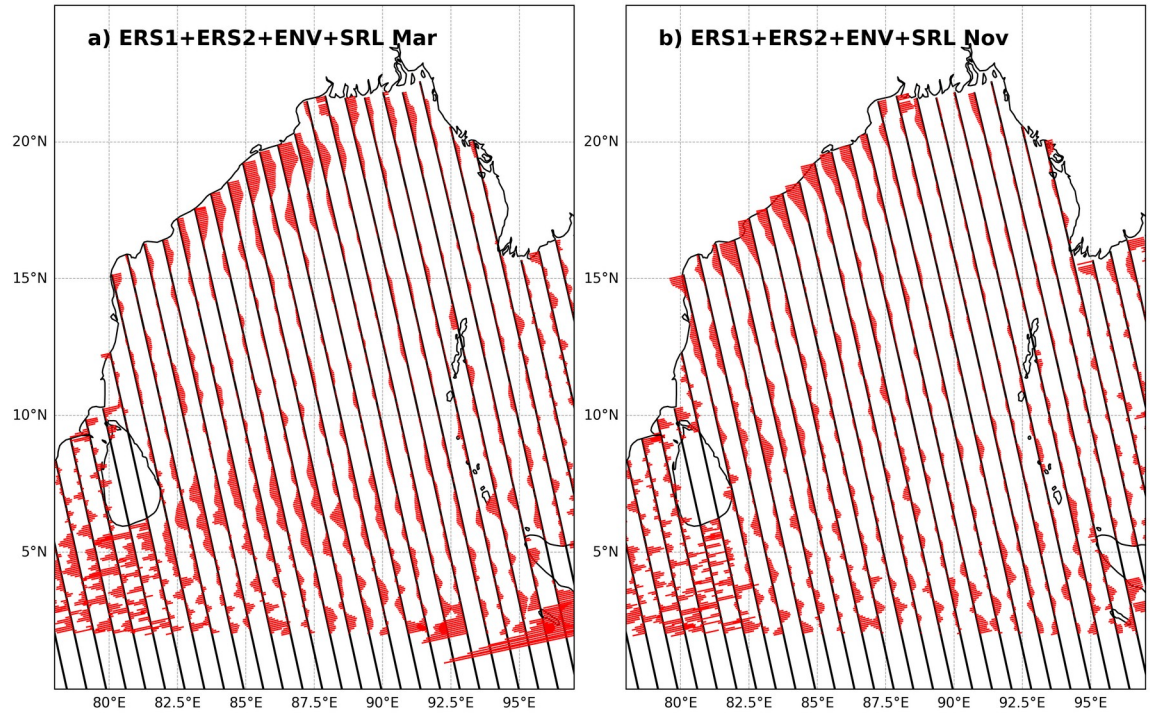


Fig6: Climatology of cross-track geostrophic currents for the months Mar (a) and Nov (b) derived from the satellite mission ERS1+ERS2+ENV+SRL. Only descending tracks are shown since cross track direction in this case makes small angle to the slope of the local coast. Thus the cross track geostrophic velocity is approximately along shelf.

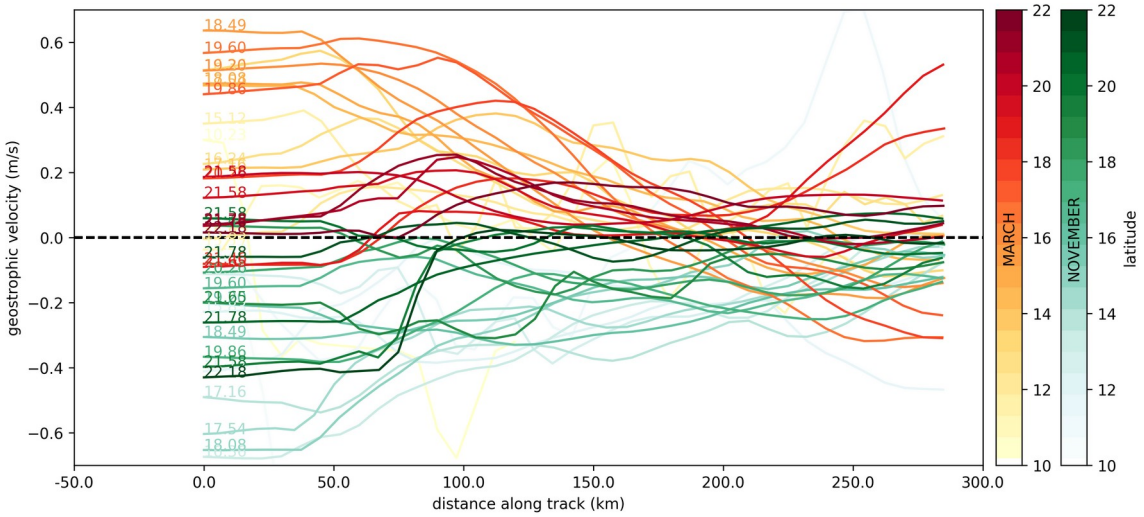


Fig7: Climatological mean cross-track velocities across the width of EICC for the months March and November at different latitudes.

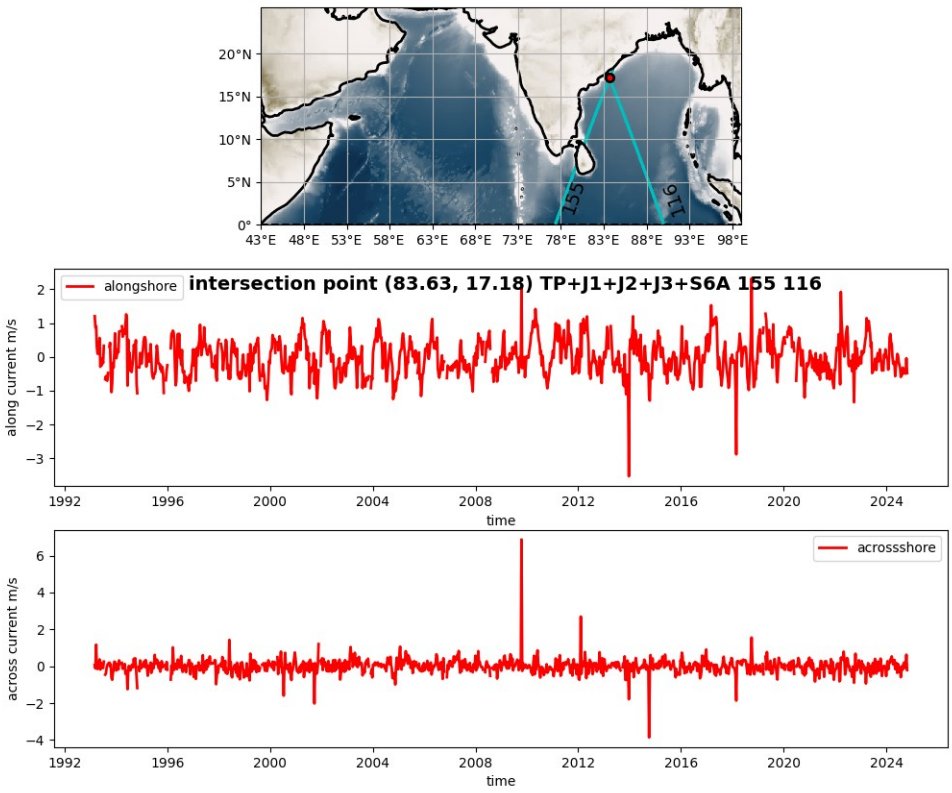


Fig8: a) An intersection point very close to the east coast of India belonging to the satellite mission TP+J1+J2+J3+S6A. b) and c) show thirty year timeseries of along shore and across shore current components at the intersection of two satellite tracks (155, 116).